

CBSE CLASS 11 MATHEMATICS

Formula & Concept Bible

2025–26 · Official CBSE Syllabus · Extensive Concepts

5 Units · 14 Chapters · 250+ Formulas · All Theorems · Conic Sections · Calculus Intro

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Official Syllabus Overview 2025-26 (Theory = 80 Marks + 20 Internal)

Unit	Chapters	Marks
I: Sets & Functions	Ch1: Sets Ch2: Relations & Functions Ch3: Trigonometric Functions	23
II: Algebra	Ch4: Complex Numbers Ch5: Linear Inequalities Ch6: P & C Ch7: Binomial Theorem Ch8: Sequences & Series	25
III: Coordinate Geometry	Ch9: Straight Lines Ch10: Conic Sections Ch11: 3D Geometry	12
IV: Calculus	Ch12: Limits and Derivatives	08
V: Statistics & Probability	Ch13: Statistics Ch14: Probability	12
	TOTAL THEORY	80



How to Use This Book

- Each chapter: Concept Summary → Visual Graph → Theorems/Proofs → Formula Reference → Common Mistakes → Worked Example → 5 Practice Questions
-  Important = high-exam-probability formulas and theorems
-  Mistake boxes = errors that cost marks in SA/LA questions every year
-  Concept boxes = proof methods and derivation strategies
-  Theorem boxes = formal statement + proof outline
- Visit sjmaths.com for free video solutions, notes, and past paper practice



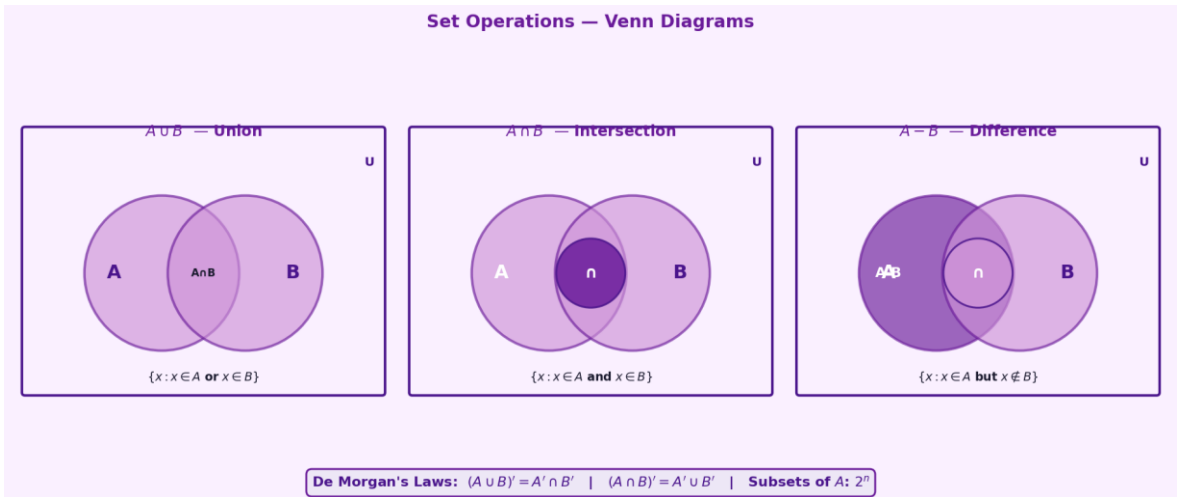
CHAPTER 1 · Unit I — Part of 23 Marks

Sets

Set Builder | Operations | Venn Diagrams | De Morgan's Laws

1.1 Representation of Sets

Method	Description	Example
Roster/Tabular	List all elements inside { }	{1, 2, 3, 4, 5}
Set-Builder	{x : property of x}	{x : x is a natural number < 6}
Venn Diagram	Circle(s) inside a rectangle (universal set)	Visual representation



Graph 1.1: Venn Diagrams — Union $A \cup B$ (left), Intersection $A \cap B$ (centre), Difference $A - B$ (right)

1.2 Types of Sets

Type	Definition	Example
Empty/Null set ($\emptyset$)	No elements $n(\emptyset)=0$	{ } or $\emptyset$
Singleton	Exactly one element	{5}
Finite	Countable number of elements	{1,2,3,...,100}
Infinite	Uncountable elements	{x : x is a real number}
Universal (U)	Contains all elements under consideration	All students in the class
Subset ($A \subseteq B$)	Every element of A is in B	$\{1,2\} \subseteq \{1,2,3\}$
Proper Subset ($A \subset B$)	$A \subseteq B$ and $A \neq B$	$\{1,2\} \subset \{1,2,3\}$

Power Set $P(A)$	Set of all subsets of A	If $A=\{1,2\}$: $P(A)=\{\emptyset,\{1\},\{2\},\{1,2\}\}$
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Number of Subsets

- If $n(A) = n$, then number of subsets = 2^n
- Number of proper subsets = $2^n - 1$
- If $n(A)=3$: $2^3=8$ subsets including $\emptyset$ and A itself

1.3 Set Operations & Laws

Operation	Symbol	Definition	Key Property
Union	$A \cup B$	$\{x: x \in A \text{ or } x \in B\}$	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Intersection	$A \cap B$	$\{x: x \in A \text{ and } x \in B\}$	$A \cap A = A \mid A \cap \emptyset = \emptyset$
Complement	A' or A^c	$\{x: x \in U \text{ but } x \notin A\}$	$A \cup A' = U \mid A \cap A' = \emptyset$
Difference	$A - B$	$\{x: x \in A \text{ but } x \notin B\}$	$A - B = A \cap B'$
Symmetric Diff.	$A \Delta B$	$(A - B) \cup (B - A)$	$= (A \cup B) - (A \cap B)$

De Morgan's Laws ★

- $(A \cup B)' = A' \cap B'$ (Complement of union = intersection of complements)
- $(A \cap B)' = A' \cup B'$ (Complement of intersection = union of complements)

- ▶ Commutative: $A \cup B = B \cup A \mid A \cap B = B \cap A$
- ▶ Associative: $(A \cup B) \cup C = A \cup (B \cup C) \mid$ same for $\cap$
- ▶ Distributive: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \mid A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

1.4 Cardinality Formulas

- ▶ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ [Inclusion – Exclusion for 2 sets]
- ▶ $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$ [for 3 sets]

Common Mistakes

- $A \subseteq B$ does NOT mean $A \subset B$. Proper subset means $A \neq B$ as well.
- $\emptyset$ (empty set) IS a subset of every set, but it is not an element unless stated.
- $n(P(A)) = 2^n$, NOT $2n$. With $n=4$, that's 16 subsets, not 8.
- In cardinality formula: always subtract $n(A \cap B)$, people often add it.

➡ Worked Example

Q: In a class of 70 students, 35 play cricket, 30 play football and 12 play both. How many play at least one sport?

Step 1: $n(C)=35$, $n(F)=30$, $n(C \cap F)=12$

Step 2: $n(C \cup F) = n(C) + n(F) - n(C \cap F) = 35 + 30 - 12 = 53$

∴ Answer: 53 students play at least one sport

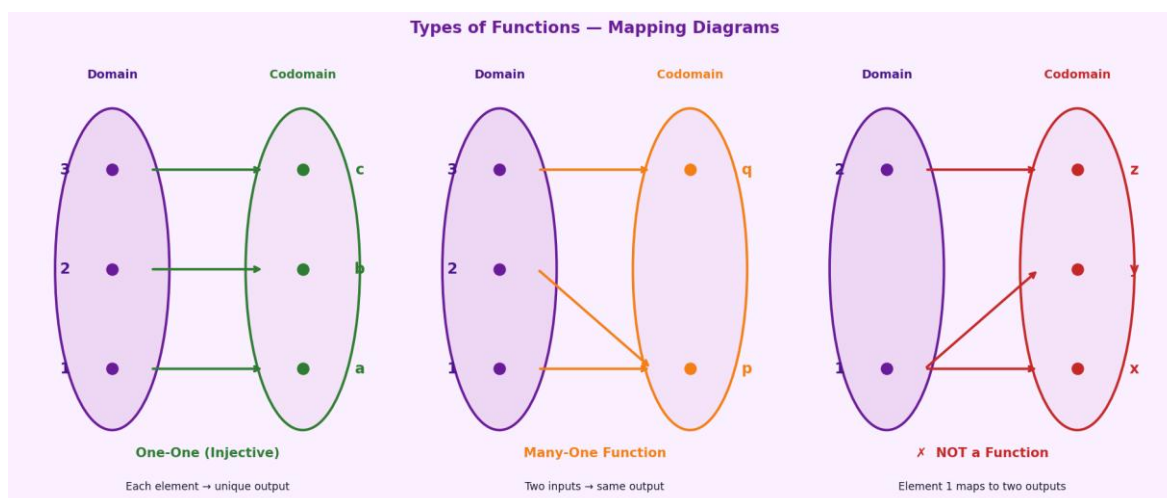
1.5 Practice Questions

Type/Year	Question
SA	If $A=\{1,2,3,4,5\}$, $B=\{2,4,6,8\}$, find $A \cup B$, $A \cap B$, and $A - B$.
SA	Verify De Morgan's law $(A \cup B)' = A' \cap B'$ for $U=\{1..10\}$, $A=\{1,3,5\}$, $B=\{3,6,9\}$.
LA	In a survey: 60% like tea, 50% like coffee, 20% like both. Find % liking at least one.
SA	If $n(A)=5$, how many subsets does A have? How many are proper subsets?
SA	Show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ using a Venn Diagram.

CHAPTER 2 · Unit I — Part of 23 Marks

Relations and Functions

Cartesian Products | Domain | Range | Types of Functions



Graph 2.1: One-one function (left), Many-one function (centre), NOT a function (right — one element maps to two)

2.1 Cartesian Product

- ▶ $A \times B = \{(a,b): a \in A, b \in B\}$ (ordered pairs)
- ▶ $n(A \times B) = n(A) \times n(B)$
- ▶ $A \times B \neq B \times A$ (unless $A=B$)
- ▶ $A \times (B \cup C) = (A \times B) \cup (A \times C)$

2.2 Relations

- ▶ A relation R from A to B is a subset of $A \times B$
- ▶ Domain of R : set of first elements | Range of R : set of second elements
- ▶ A function is a special relation where each element of domain has exactly ONE image

2.3 Functions — Key Concepts

Type	Condition	Example
One-one (Injective)	$f(a)=f(b) \Rightarrow a=b$ (distinct inputs give distinct outputs)	$f(x)=2x$
Onto (Surjective)	Range = Codomain (every output has a preimage)	$f: \mathbb{R} \rightarrow \mathbb{R}, f(x)=2x+1$
Bijjective	One-one AND onto	$f(x)=x$
Into	Range $\subseteq$ Codomain (not necessarily equal)	$f: \{1,2\} \rightarrow \{1,2,3\}, f(x)=x$

Many-one	Two or more inputs give same output	$f(x)=x^2$
Constant	$f(x)=c$ for all x	$f(x)=5$

2.4 Domain, Range & Algebra of Functions

- ▶ For $f(x)+g(x)$: Domain = $\text{Dom}(f) \cap \text{Dom}(g)$
- ▶ For $f(x)/g(x)$: Domain = $\text{Dom}(f) \cap \text{Dom}(g)$ excluding points where $g(x)=0$
- ▶ Range: set of all actual output values produced by the function
- ▶ Real functions domain: exclude values making denominator 0 or square root of negative

Standard Domain/Range

- $f(x)=\sqrt{x}$: Domain= $[0, \infty)$, Range= $[0, \infty)$
- $f(x)=1/x$: Domain= $\mathbb{R}-\{0\}$, Range= $\mathbb{R}-\{0\}$
- $f(x)=|x|$: Domain= $\mathbb{R}$, Range= $[0, \infty)$
- $f(x)=x^2$: Domain= $\mathbb{R}$, Range= $[0, \infty)$
- Signum function: $\text{sgn}(x)=1$ if $x>0$, 0 if $x=0$, -1 if $x<0$

Common Mistakes

- Not every relation is a function. Check: each x in domain has unique y .
- Domain is NOT always $\mathbb{R}$. Always check for restrictions (zero denominator, negative under root).
- $f \circ g$ means $f(g(x))$: apply g FIRST, then f . Order matters!
- Equal functions must have same domain AND same rule.

Worked Example

Q: Find the domain and range of $f(x) = 1/(x^2-4)$

Step 1: Denominator $x^2-4=0 \Rightarrow x=\pm 2$. These must be excluded.

Step 2: Domain = $\mathbb{R} - \{2, -2\}$

Step 3: As $x \rightarrow \pm 2$, $f(x) \rightarrow \pm \infty$ | As $x \rightarrow \infty$, $f(x) \rightarrow 0^+$. $f(x)$ can be any real except 0.

Step 4: Range = $\mathbb{R} - \{0\}$

∴ Answer: Domain = $\mathbb{R}-\{\pm 2\}$, Range = $\mathbb{R}-\{0\}$

2.5 Practice Questions

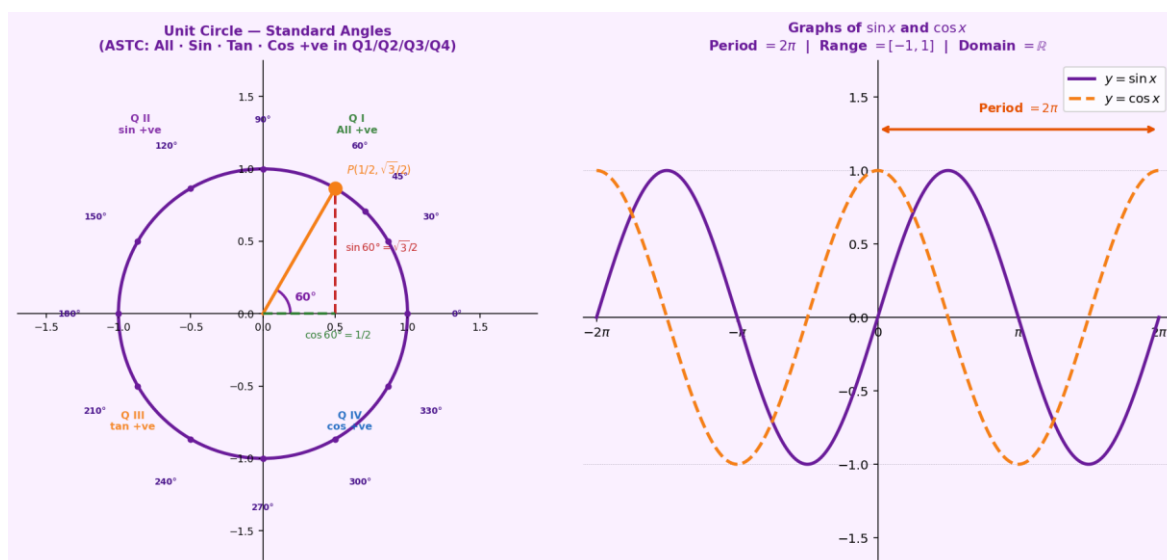
Type/Year	Question
SA	If $A=\{1,2,3\}$ and $B=\{a,b\}$, find $A \times B$ and $n(A \times B)$.
SA	Find domain and range of $f(x) = \sqrt{9-x^2}$.
LA	Let $f(x)=x^2$ and $g(x)=2x+1$. Find $f \circ g$ and $g \circ f$. Are they equal?
SA	Determine if $f(x)= x $ is one-one and/or onto when $f:\mathbb{R} \rightarrow \mathbb{R}$.

SAFind the domain of $f(x) = 1/\sqrt{(x^2-5x+6)}$.

CHAPTER 3 · Unit I — Part of 23 Marks

Trigonometric Functions

Radian Measure | All Ratios | Identities | Graphs | Equations

Graph 3.1: Unit circle with standard angles (left) and graphs of $\sin x$ and $\cos x$ with period 2π (right)

3.1 Angle Measurement

- **Radian↔Degree:** θ (radians) = $\theta \times \frac{180}{\pi}$ degrees | θ (degrees) = $\theta \times \frac{\pi}{180}$ radians
- Arc length: $l = r\theta$ (r = radius, θ in radians)
- Area of sector: $A = \frac{1}{2} r^2 \theta$

3.2 Standard Values Table

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	Undefined
$\operatorname{cosec} \theta$	Undefined	2	$\sqrt{2}$	$2/\sqrt{3}$	1
$\sec \theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	Undefined
$\cot \theta$	Undefined	$\sqrt{3}$	1	$1/\sqrt{3}$	0

3.3 Fundamental Identities

- $\sin^2 \theta + \cos^2 \theta = 1$

- ▶ $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \sec^2\theta = 1 + \tan^2\theta$
- ▶ $\operatorname{cosec}^2\theta - \cot^2\theta = 1 \Rightarrow \operatorname{cosec}^2\theta = 1 + \cot^2\theta$

3.4 Compound Angle Formulas ★

- ▶ $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
- ▶ $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- ▶ $\tan(A \pm B) = (\tan A \pm \tan B) / (1 \mp \tan A \tan B)$
- ▶ $\sin 2A = 2 \sin A \cos A = 2 \tan A / (1 + \tan^2 A)$
- ▶ $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1 = (1 - \tan^2 A) / (1 + \tan^2 A)$
- ▶ $\tan 2A = 2 \tan A / (1 - \tan^2 A)$
- ▶ $\sin 3A = 3 \sin A - 4 \sin^3 A \quad | \quad \cos 3A = 4 \cos^3 A - 3 \cos A$

3.5 Product-to-Sum & Sum-to-Product

- ▶ $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
- ▶ $2 \cos A \cos B = \cos(A - B) + \cos(A + B)$
- ▶ $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
- ▶ $\sin C + \sin D = 2 \sin((C + D)/2) \cos((C - D)/2)$
- ▶ $\sin C - \sin D = 2 \cos((C + D)/2) \sin((C - D)/2)$
- ▶ $\cos C + \cos D = 2 \cos((C + D)/2) \cos((C - D)/2)$
- ▶ $\cos C - \cos D = -2 \sin((C + D)/2) \sin((C - D)/2)$

3.6 Signs & ASTC in Four Quadrants

Quadrant	Angle Range	Positive Ratios
I	0° to 90°	All (sin, cos, tan, sec, cosec, cot)
II	90° to 180°	sin and cosec only
III	180° to 270°	tan and cot only
IV	270° to 360°	cos and sec only

💡 Mnemonic: ASTC

- All Students Take Calculus — All / Sin / Tan / Cos are positive in Q1/Q2/Q3/Q4 respectively
- Complementary: $\sin(90 - \theta) = \cos \theta$ | $\cos(90 - \theta) = \sin \theta$ | $\tan(90 - \theta) = \cot \theta$
- Supplementary: $\sin(180 - \theta) = \sin \theta$ | $\cos(180 - \theta) = -\cos \theta$

3.7 Graphs: Period & Range

Function	Period	Domain	Range
sin x	2π	R	$[-1, 1]$
cos x	2π	R	$[-1, 1]$
tan x	π	$R - \{(2n+1)\pi/2\}$	R

cosec x	2π	$R - \{n\pi\}$	$(-\infty, -1] \cup [1, \infty)$
sec x	2π	$R - \{(2n+1)\pi/2\}$	$(-\infty, -1] \cup [1, \infty)$
cot x	π	$R - \{n\pi\}$	R

3.8 Trigonometric Equations

- ▶ General solution of $\sin\theta = 0$: $\theta = n\pi$, $n \in \mathbb{Z}$
- ▶ General solution of $\cos\theta = 0$: $\theta = (2n+1)\pi/2$, $n \in \mathbb{Z}$
- ▶ General solution of $\tan\theta = 0$: $\theta = n\pi$, $n \in \mathbb{Z}$
- ▶ General solution of $\sin\theta = \sin\alpha$: $\theta = n\pi + (-1)^n\alpha$, $n \in \mathbb{Z}$
- ▶ General solution of $\cos\theta = \cos\alpha$: $\theta = 2n\pi \pm \alpha$, $n \in \mathbb{Z}$
- ▶ General solution of $\tan\theta = \tan\alpha$: $\theta = n\pi + \alpha$, $n \in \mathbb{Z}$

⚠ Common Mistakes

- $\sin(A+B) \neq \sin A + \sin B$. Always use the compound angle formula.
- $\tan 90^\circ$ is UNDEFINED, not infinity. Never substitute 90° into $\tan$.
- $\cos 2A$ has THREE equivalent forms. Choose the one that simplifies the expression.
- ASTC: in Q2, only $\sin$ and $\csc$ are positive. $\cos$, $\tan$, $\cot$, $\sec$ are negative.

➡ Worked Example

Q: Prove: $(\sin 3A + \sin A)/(\cos 3A + \cos A) = \tan 2A$

Step 1: Numerator: $\sin 3A + \sin A = 2\sin(2A)\cos(A)$ [sum-to-product]

Step 2: Denominator: $\cos 3A + \cos A = 2\cos(2A)\cos(A)$ [sum-to-product]

Step 3: Ratio = $2\sin 2A \cos A / (2\cos 2A \cos A) = \sin 2A / \cos 2A = \tan 2A$

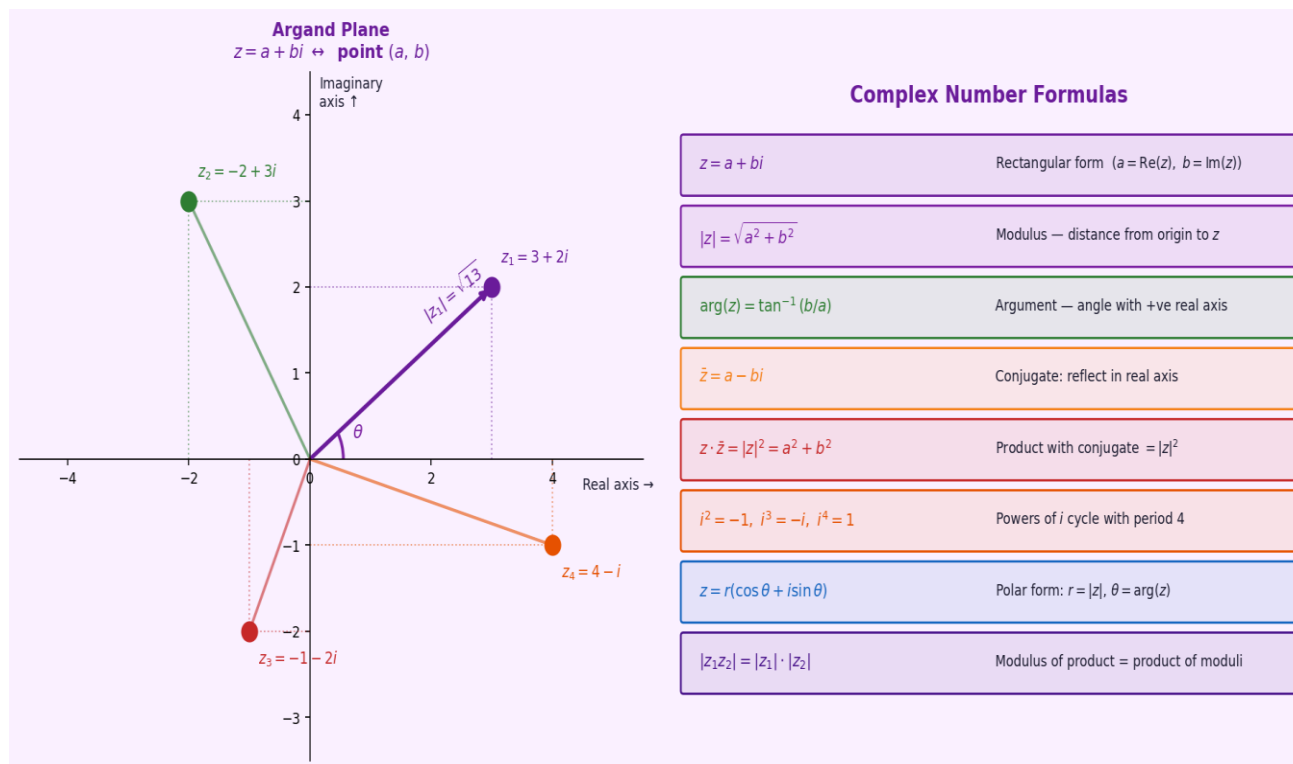
∴ Answer: LHS = $\tan 2A$ = RHS (proved)

Type/Year	Question
SA	Prove: $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$
SA	Find all values of x where $2\cos^2 x + 3\sin x = 0$, $x \in [0, 2\pi]$.
LA	Prove: $\cos^2 x + \cos^2(x+\pi/3) + \cos^2(x-\pi/3) = 3/2$
SA	Express $\sin 75^\circ$ in exact form using compound angle formula.
SA	Find the general solution of $\sqrt{3} \tan x = 1$.

CHAPTER 4 · Unit II — Part of 25 Marks

Complex Numbers and Quadratic Equations

Argand Plane | Modulus | Argument | Polar Form



Graph 4.1: Argand Plane with 4 plotted complex numbers and modulus vector (left) | Key properties reference (right)

4.1 Complex Number Basics

- ▶ $z = a + bi$ where $a = \text{Real part } (\text{Re}(z))$, $b = \text{Imaginary part } (\text{Im}(z))$, $i = \sqrt{-1}$
- ▶ $i^1 = i$ | $i^2 = -1$ | $i^3 = -i$ | $i^4 = 1$ (cycle of 4, so $i^n = i^{(n \bmod 4)}$)
- ▶ Two complex numbers equal: $a + bi = c + di \Leftrightarrow a = c$ AND $b = d$

4.2 Arithmetic Operations

Operation	Formula
Addition	$(a+bi) + (c+di) = (a+c) + (b+d)i$
Subtraction	$(a+bi) - (c+di) = (a-c) + (b-d)i$
Multiplication	$(a+bi)(c+di) = (ac-bd) + (ad+bc)i$ [use $i^2 = -1$]
Division	Multiply numerator and denominator by conjugate of denominator
Conjugate	$\bar{z} = a - bi$ $z + \bar{z} = 2a$ (real) $z - \bar{z} = 2bi$ (pure imaginary)
$z \times \bar{z}$	$= a^2 + b^2 = z ^2$

4.3 Modulus and Argument

- ▶ $|z| = \sqrt{a^2 + b^2}$ *always* ≥ 0
- ▶ $\arg(z) = \theta = \tan^{-1}(b/a)$ [adjust for correct quadrant!]
- ▶ Polar form: $z = r(\cos\theta + i \sin\theta)$ where $r=|z|$, $\theta=\arg(z)$
- ▶ Triangle inequality: $|z_1+z_2| \leq |z_1|+|z_2|$
- ▶ $|z_1 \times z_2| = |z_1| \times |z_2|$ | $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$

4.4 Quadratic Equations with Complex Roots

- ▶ If $b^2-4ac < 0$: roots are complex conjugates $x = [-b \pm i\sqrt{(4ac-b^2)}] / 2a$
- ▶ Complex roots always occur in conjugate pairs in equations with real coefficients

Key Properties of Modulus

- $|z| = 0 \Leftrightarrow z = 0$ (the only complex number with modulus 0)
- $|z| = |\bar{z}|$ (conjugate has same modulus)
- $|z_1 z_2| = |z_1| |z_2|$ | $|z_1/z_2| = |z_1|/|z_2|$
- $\operatorname{Re}(z) \leq |z|$ and $\operatorname{Im}(z) \leq |z|$ (always true)

Common Mistakes

- $i^2 = -1$ NOT $+1$. When expanding $(a+bi)(c+di)$, $bi \times di = bdi^2 = -bd$ (not $+bd$).
- $\arg(z)$ is NOT simply $\tan^{-1}(b/a)$. Adjust based on quadrant of (a,b) .
- Division: always multiply by conjugate of DENOMINATOR, not numerator.
- $\sqrt{-4} \neq \pm 2$. Correctly: $\sqrt{-4} = 2i$.

Worked Example

Q: Express $(3+2i)/(1-i)$ in the form $a+bi$ and find its modulus.

Step 1: Multiply by conjugate $(1+i)/(1+i)$: $= (3+2i)(1+i)/((1-i)(1+i))$

Step 2: Numerator: $(3+2i)(1+i) = 3+3i+2i+2i^2 = 3+5i-2 = 1+5i$

Step 3: Denominator: $(1-i)(1+i) = 1-i^2 = 1+1 = 2$

Step 4: Result: $(1+5i)/2 = 1/2 + 5i/2$

Step 5: $|z| = \sqrt{((1/2)^2 + (5/2)^2)} = \sqrt{(1/4 + 25/4)} = \sqrt{(26/4)} = \sqrt{26}/2$

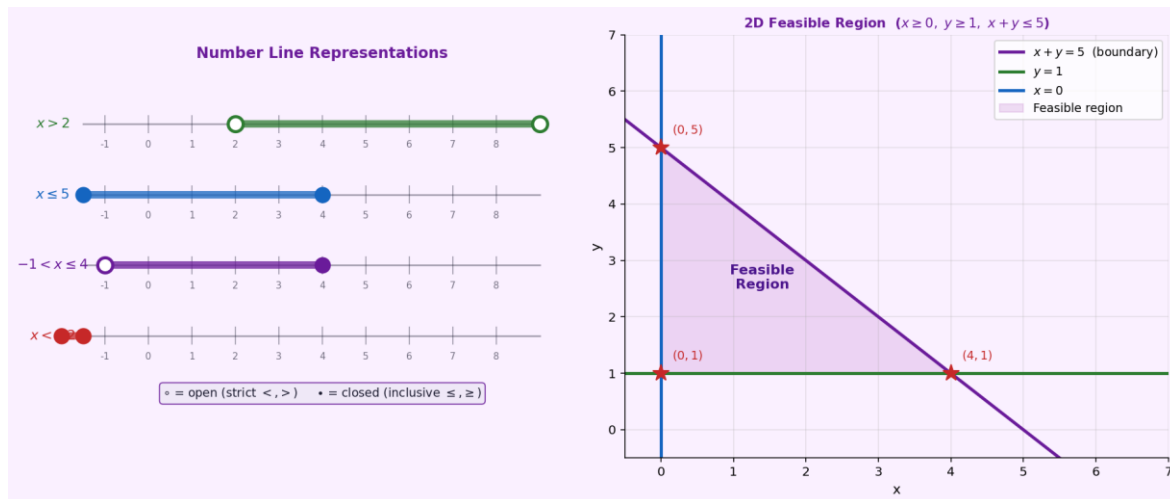
∴ Answer: $z = 1/2 + 5i/2$, $|z| = \sqrt{26}/2$

Type/Year	Question
SA	Find real values of x and y: $(x+iy)(2-i) = 3+i$.
SA	Express $1/(1+\cos\theta+i \sin\theta)$ in the form $a+bi$.
SA	If $ z =2$, find the maximum and minimum values of $ z+1/z $.
LA	Solve: $x^2-2x+5=0$ over complex numbers. Verify roots are conjugates.
SA	Show that $ (z_1+z_2) ^2 + (z_1-z_2) ^2 = 2(z_1 ^2 + z_2 ^2)$.

CHAPTER 5 · Unit II — Part of 25 Marks

Linear Inequalities

Number Line | 2D Feasible Region | Absolute Value



Graph 5.1: Number line representations of inequalities (left) and 2D feasible region (right)

5.1 Fundamental Rules for Inequalities

Rules for Manipulating Inequalities

- Add/subtract same number to both sides: inequality UNCHANGED
- Multiply/divide by POSITIVE number: inequality UNCHANGED
- Multiply/divide by NEGATIVE number: inequality REVERSES direction
- If $a < b$ and $b < c$ then $a < c$ (transitivity)
- Never multiply/divide by a variable unless you know its sign!

5.2 Solving Linear Inequalities in One Variable

- ▶ Bring all terms to one side, simplify, find solution set, represent on number line
- ▶ Open circle $\circ$: strict inequality ($<$ or $>$) | Filled circle $\bullet$: non-strict ($\leq$ or $\geq$)
- ▶ Solution in interval notation: $x > 2 \Rightarrow x \in (2, \infty)$ | $x \leq 5 \Rightarrow x \in (-\infty, 5]$
- ▶ $-2 < x \leq 3 \Rightarrow x \in (-2, 3]$ (left open, right closed)

5.3 Linear Inequalities in Two Variables

Graphing Method

- Step 1: Replace inequality with $=$, draw the boundary line
- Step 2: If inequality is strict ($<$ or $>$): use dashed line; if $\leq$ or $\geq$: use solid line
- Step 3: Test a point not on the line (origin $(0,0)$ is easiest)

- Step 4: If test point satisfies inequality, shade that half-plane; otherwise shade other side
- Step 5: The shaded region is the solution (feasible region)

⚠ Common Mistakes

- When multiplying both sides by negative: $3 > 2$ becomes $\sqrt{3} < -2$. Direction FLIPS.
- Dividing by variable: if x is negative, direction reverses — always check or split cases.
- For $\leq$ inequalities: boundary line is included (solid line), not dotted.
- $-1 < x < 3$ does NOT mean x is between -1 and 3 exclusive. It includes all real numbers in that range.

➡ Worked Example

Q: Solve: $3x-2 \geq 5x+4$. Represent on number line.

Step 1: $3x-2 \geq 5x+4$

Step 2: $3x-5x \geq 4+2$ (move x terms to left, constants to right)

Step 3: $-2x \geq 6$

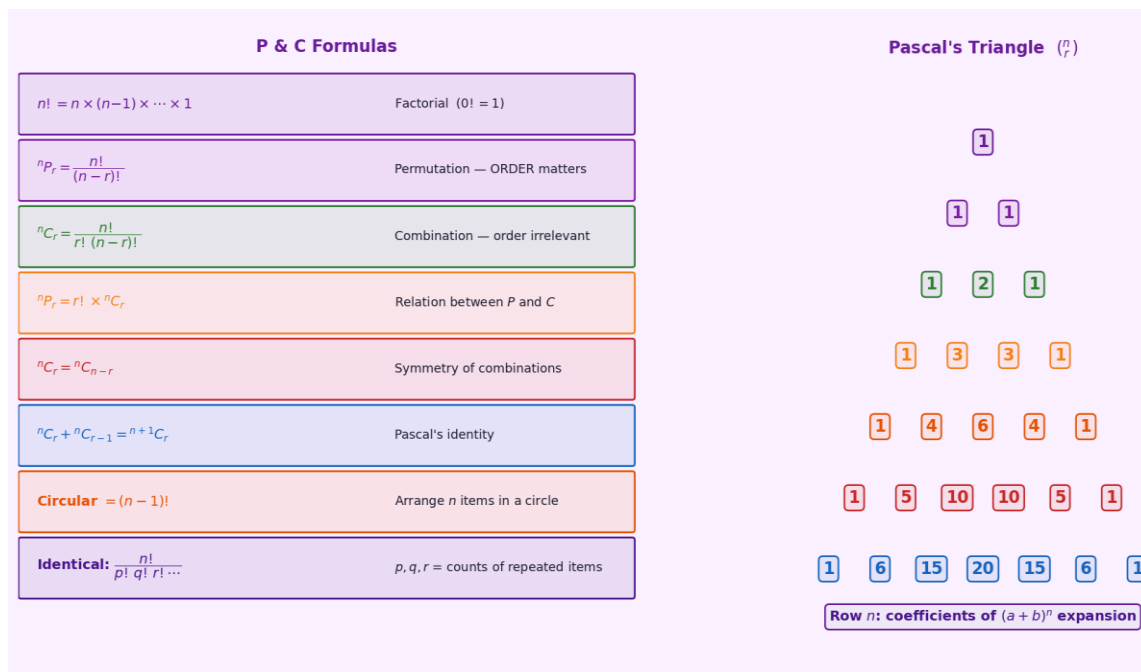
Step 4: Divide by -2 (reverse inequality): $x \leq -3$

∴ Answer: $x \leq -3$, i.e., $x \in (-\infty, -3]$

Type/Year	Question
SA	Solve: $3(x-1) \leq 2(x-3)$. Represent the solution on a number line.
SA	Solve and graph: $ 2x+3 < 7$.
LA	Graph the system: $x+y>4$, $x-1<0$, $y<4$. Find vertices of feasible region.
SA	A student scores 60, 72, 48, 80 in 4 tests. What range must the 5th score be to average > 65 ?
SA	Solve: $(x-1)/(x+3) > 0$.

CHAPTER 6 · Unit II — Part of 25 Marks

Permutations and Combinations

Fundamental Principle | Factorial | nPr | nCr Graph 6.1: P&C formula reference (left) and Pascal's Triangle showing nCr coefficients (right)

6.1 Fundamental Principles of Counting



Multiplication Principle

- If one task can be done in m ways and a second in n ways (independently), total ways = $m \times n$
- Example: 3 shirts and 4 trousers $\Rightarrow 3 \times 4 = 12$ outfits



Addition Principle

- If one task can be done in m ways OR another in n ways (mutually exclusive), total = $m + n$
- Example: 3 buses or 4 trains $\Rightarrow 3 + 4 = 7$ ways to travel

6.2 Factorial Notation

- $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$
- $0! = 1$ and $1! = 1$ (by definition)
- $n! = n \times (n-1)! \Rightarrow$ Use recursion to compute: $5! = 5 \times 4! = 5 \times 24 = 120$



6.3 Permutations nPr

- ▶ $nPr = \frac{n!}{(n-r)!}$ *number of ways to arrange r items from n (ORDER matters)*
- ▶ $nPn = n!$ (arrange all n items)
- ▶ $nP_1 = n$ | $nP_2 = n(n-1)$ | $nP_0 = 1$

6.4 Permutations with Repetition

- ▶ Arrangements of n items with some repeated: $n!/(p!q!r!...)$ where $p, q, r, ...$ are frequencies of repeated items
- ▶ Circular permutations of n distinct items = $(n-1)!$
- ▶ If n items can repeat: n^r ways to arrange r positions

6.5 Combinations nCr

- ▶ $nCr = \frac{n!}{r!(n-r)!}$ *number of ways to choose r from n (ORDER doesn't matter)*
- ▶ **Key identities:** $nCr = nC_{(n-r)}$ | $nC_0 = nC_n = 1$ | $nCr + nC_{r-1} = (n-1)C_r$ [Pascal]



P vs C Decision Guide

- Use PERMUTATION (nPr) when ORDER matters: passwords, rankings, seating in a row
- Use COMBINATION (nCr) when ORDER doesn't matter: selecting team, choosing topics
- $nPr = r! \times nCr$ (relating the two)
- nCr is always a whole number

⚠ Common Mistakes

- $0! = 1$ (not 0). This is a definition, not a calculation.
- In circular arrangements: $(n-1)!$, NOT $n!$. One position is fixed as reference.
- Identical objects: REDUCE by dividing by their factorials. AABB arrangement: $4!/(2!2!)=6$.
- If question says 'at least': use complementary counting = Total – (none/too few).

➡ Worked Example

Q: How many 4-digit numbers can be formed from digits 1,2,3,4,5 if (a) repetition allowed (b) not allowed?

Step 1: (a) Each digit can be any of 5. Total = $5 \times 5 \times 5 \times 5 = 5^4 = 625$

Step 2: (b) Arrange 4 from 5 without repetition: ${}^5P_4 = 5!/(5-4)! = 5 \times 4 \times 3 \times 2 = 120$

∴ Answer: (a) 625 (b) 120

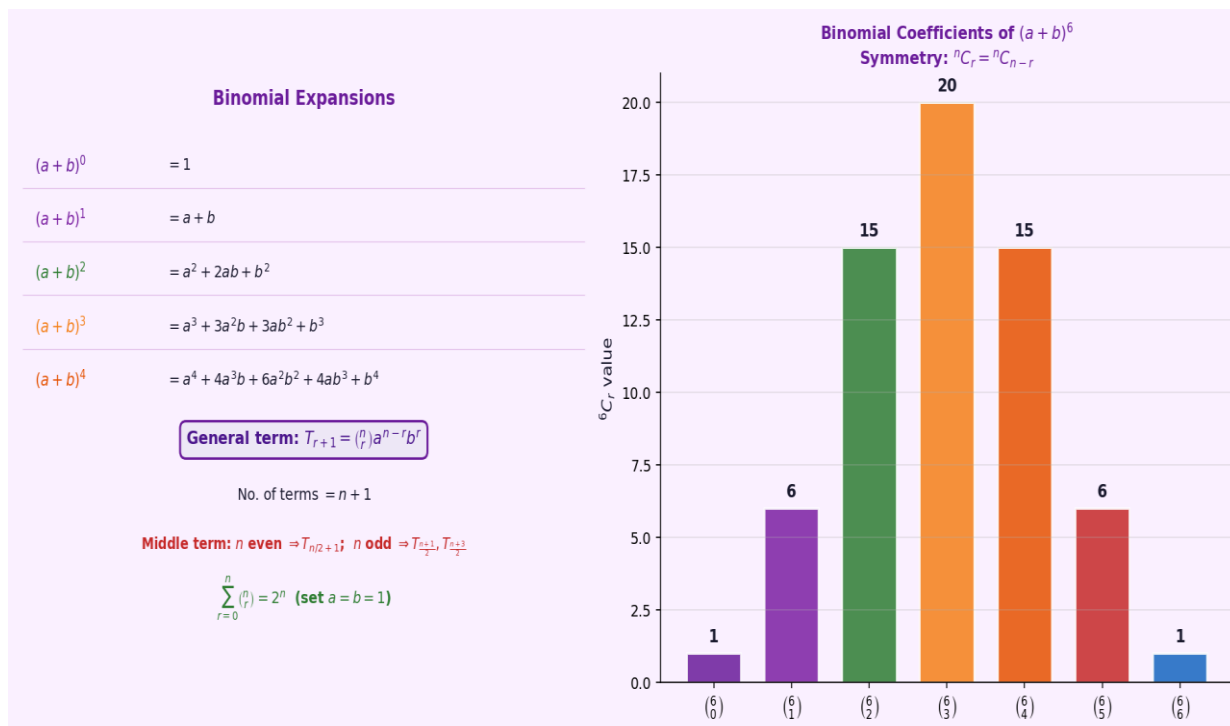
Type/Year	Question
SA	How many ways can 5 students sit in a row? In a circle?
SA	In how many ways can a committee of 3 be chosen from 8 people?
LA	Find the number of 5-letter words using letters of INDIA with no repetition.



SA	How many diagonals does a hexagon have? (Use $nCr - n$ sides)
SA	Verify: ${}^{10}C_3 + {}^{10}C_4 = {}^{11}C_4$ (Pascal's identity)

CHAPTER 7 · Unit II — Part of 25 Marks

Binomial Theorem

General Term | Middle Term | nCr Coefficients | Pascal's TriangleGraph 7.1: Binomial expansions for $n=0$ to 4 (left) and bar chart of nC_r coefficients showing symmetry (right)

7.1 The Binomial Theorem

**Binomial Theorem (key statement)**

- $(a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_n a^0 b^n$
- $= \sum {}^nC_r a^{n-r} b^r$ (sum from $r=0$ to n)
- Total number of terms in expansion $= n + 1$
- Coefficients (${}^nC_0, {}^nC_1, \dots, {}^nC_n$) are called binomial coefficients

7.2 General Term

- ▶ **General term $T(r+1) = {}^nC_r \times a^{n-r} \times b^r$ $r = 0, 1, 2, \dots, n$**
- ▶ To find a specific term: set exponent of one variable equal to r and solve
- ▶ Term containing b^k : set exponent of $b = k$, that term is $T(k+1)$



7.3 Middle Term(s)

💡 Finding Middle Terms

- If n is EVEN: only one middle term = $T(n/2 + 1)$ [e.g., $n=6 \Rightarrow T(4)$]
- If n is ODD: TWO middle terms = $T((n+1)/2)$ and $T((n+3)/2)$ [e.g., $n=5 \Rightarrow T(3)$ and $T(4)$]
- r starts at 0, so $T(1)$ is the first term (not $T(0)$)

7.4 Properties of Binomial Coefficients

- ▶ ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$ (substitute $a=b=1$)
- ▶ ${}^nC_0 - {}^nC_1 + {}^nC_2 - \dots = 0$ (substitute $a=1, b=-1$)
- ▶ ${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = {}^nC_1 + {}^nC_3 + \dots = 2^{n-1}$
- ▶ ${}^nC_r = {}^nC_{n-r}$ (symmetry property: used to simplify sums)

⚠ Common Mistakes

- Number of terms = $n+1$, NOT n . Expansion of $(a+b)^3$ has 4 terms.
- In $T(r+1) = {}^nC_r a^{n-r} b^r$: the exponents of a and b sum to n always.
- Middle term: $n=6$ gives $T(4)$, $n=7$ gives $T(4)$ AND $T(5)$. Even/odd matters!
- When expanding $(1+x)^n$: $a=1, b=x$. Don't write $a=x$.

➡ Worked Example

Q: Find the 5th term in expansion of $(x+2)^{10}$.

Step 1: General term $T(r+1) = {}^{10}C_r \times x^{10-r} \times 2^r$

Step 2: 5th term $\Rightarrow r+1=5 \Rightarrow r=4$

Step 3: $T(5) = {}^{10}C_4 \times x^{10-4} \times 2^4 = 210 \times x^6 \times 16 = 3360x^6$

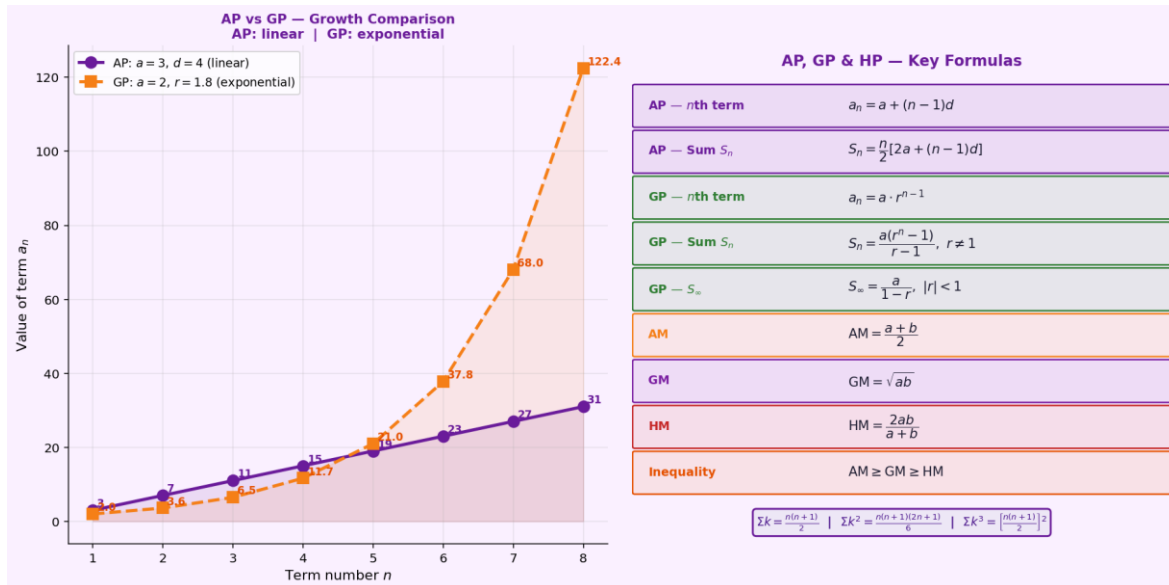
∴ Answer: $T(5) = 3360x^6$

Type/Year	Question
SA	Find the middle term of $(3x-2/x^2)^9$.
SA	Find the coefficient of x^5 in $(x+3)^8$.
SA	Prove that ${}^nC_0 + {}^nC_1 + \dots + {}^nC_n = 2^n$ by substituting appropriate values.
LA	Find the term independent of x in $(\sqrt{x}-1/(2x^2))^{10}$.
SA	If the 3rd and 4th terms in $(ax+b)^5$ have equal coefficients, find a/b .

CHAPTER 8 · Unit II — Part of 25 Marks

Sequences and Series

AP | GP | Sum Formulas | AM GM HM | Special Series



Graph 8.1: AP (linear) vs GP (exponential) growth comparison (left) | AP & GP formula reference table (right)

8.1 Arithmetic Progression (AP)

- ▶ **n th term:** $a^n = a + (n-1)d$ where a = first term, d = common difference
- ▶ **Sum of n terms:** $S^n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$ where l = last term
- ▶ Common difference $d = a_2 - a_1 = a_3 - a_2$ (constant throughout)
- ▶ If S^n is sum of n terms, then $a^n = S^n - S^{n-1}$ (for $n \geq 2$)
- ▶ Arithmetic Mean AM between a and b : $AM = (a+b)/2$
- ▶ If a, AM, b are in AP: $2AM = a+b$

8.2 Geometric Progression (GP)

- ▶ **n th term:** $a^n = a \times r^{n-1}$ *a =first term, r =common ratio*
- ▶ **Sum of n terms:** $S^n = \frac{a(r^n - 1)}{r - 1}$ ($r \neq 1$) | $S^n = na$ ($r = 1$)
- ▶ **Sum of infinite GP ($|r| < 1$):** $S^\infty = \frac{a}{1-r}$
- ▶ Common ratio $r = a_2/a_1 = a_3/a_2$ (constant throughout)
- ▶ Geometric Mean between a and b : $GM = \sqrt{ab}$ | Always: $AM \geq GM \geq HM$ (for positive numbers)

8.3 Harmonic Progression (HP)

- ▶ a, b, c are in HP if $1/a, 1/b, 1/c$ are in AP
- ▶ Harmonic Mean: $HM = 2ab/(a+b)$
- ▶ Key relation: $AM \times HM = GM^2$

8.4 Special Series Sums ★

► $\Sigma k = \frac{n(n+1)}{2}$ *Sum of first n natural numbers*

► $\Sigma k^2 = \frac{n(n+1)(2n+1)}{6}$ *Sum of squares*

► $\Sigma k^3 = \left(\frac{n(n+1)}{2}\right)^2$ *Sum of cubes = $(\Sigma k)^2$*

⚠ Common Mistakes

- $d = a_2 - a_1$ (second minus first, NOT first minus second). Negative d means decreasing AP.
- Sum of infinite GP only works when $|r| < 1$. If $|r| \geq 1$, sum diverges to infinity.
- $AM \geq GM$: $AM = (a+b)/2$, $GM = \sqrt{ab}$. Equality holds only when $a = b$.
- HP: work with the reciprocals in AP. Never directly apply AP formula to HP.

➡ Worked Example

Q: The 7th term of an AP is 15 and the 13th term is 27. Find the AP and sum of first 20 terms.

Step 1: $a_7 = a + 6d = 15$ and $a_{13} = a + 12d = 27$

Step 2: Subtracting: $6d = 12 \Rightarrow d = 2$

Step 3: $a + 6(2) = 15 \Rightarrow a = 3$

Step 4: AP: 3, 5, 7, 9, ...

Step 5: $S_{20} = 20/2 \times [2(3) + (20-1) \times 2] = 10 \times [6 + 38] = 10 \times 44 = 440$

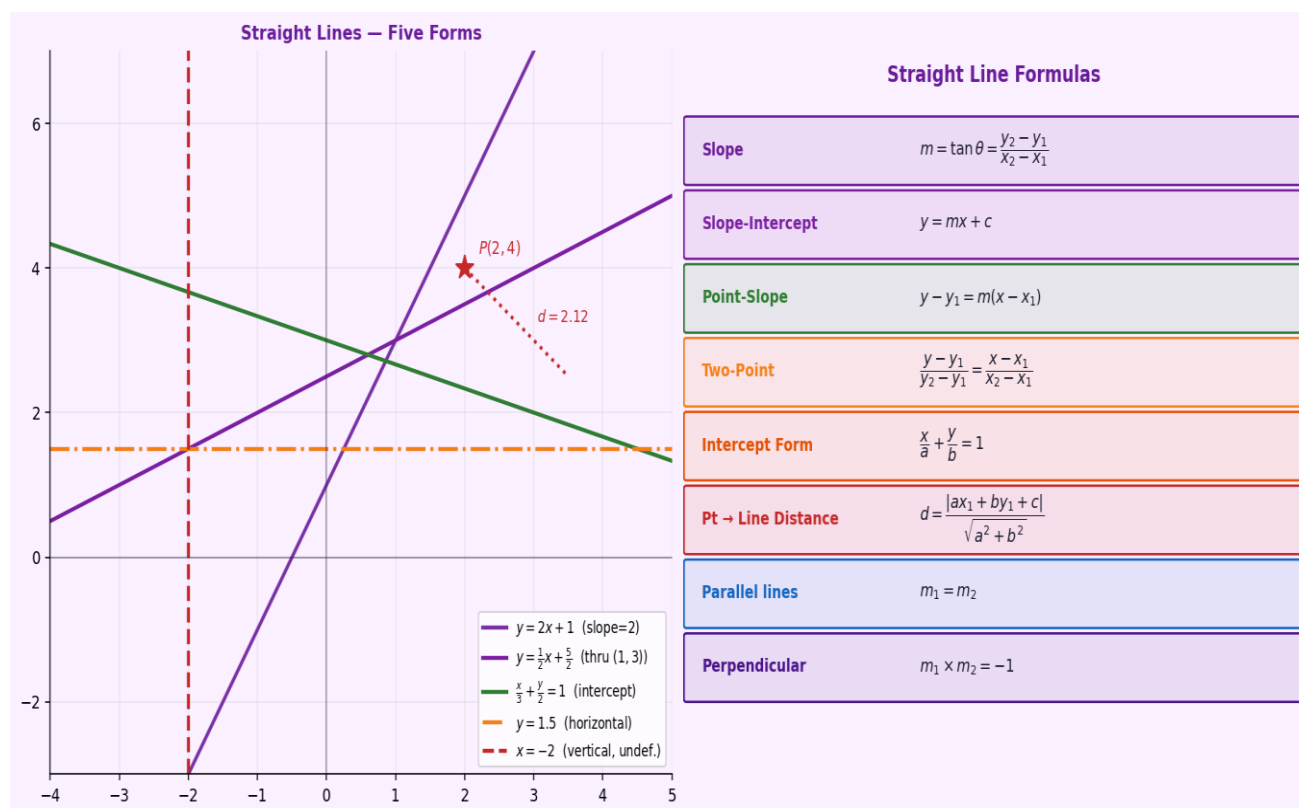
∴ Answer: $a=3$, $d=2$, $S_{20} = 440$

Type/Year	Question
SA	Find S_{10} of AP: 2, 7, 12, 17, ...
SA	Find 8th term of GP: 3, 6, 12, 24, ...
SA	Sum of infinite GP with first term 1 and ratio $1/2$. Verify: $S_{\infty} = 2$.
LA	If a, b, c are in AP and x, y, z in GP, prove: $x^{b-c} \cdot y^{c-a} \cdot z^{a-b} = 1$.
SA	Find n if $1+3+5+\dots+(2n-1) = 100$.

CHAPTER 9 · Unit III — Part of 12 Marks

Straight Lines

Slope | Multiple Forms | Angle Between Lines | Distance



Graph 9.1: Various line forms including slope-intercept, horizontal, vertical (left) | Distance from point P to line (right)

9.1 Slope (Gradient)

- ▶ $m = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$ θ = angle with positive x-axis (inclination)
- ▶ Parallel lines: $m_1 = m_2$ | Perpendicular lines: $m_1 \times m_2 = -1$
- ▶ Horizontal line: $m = 0$ | Vertical line: $m = \text{undefined}$ ($\tan 90^\circ$)

9.2 Equations of Lines — All 5 Forms

Form	Equation	When to Use
Slope-intercept	$y = mx + c$	Slope m and y-intercept c known
Point-slope	$y - y_1 = m(x - x_1)$	Slope m and one point (x_1, y_1) known
Two-point	$(y - y_1)/(y_2 - y_1) = (x - x_1)/(x_2 - x_1)$	Two points given
Intercept form	$x/a + y/b = 1$	x-intercept a and y-intercept b known
Normal form	$x \cos \omega + y \sin \omega = p$	Length p of perpendicular from origin, angle ω

9.3 Angle Between Two Lines

- ▶ $\tan\theta = \pm \frac{(m_1 - m_2)}{1 + m_1 m_2}$ $\theta = \text{acute angle between lines with slopes } m_1 \text{ and } m_2$
- ▶ Perpendicular condition: $m_1 m_2 = -1$
- ▶ Parallel condition: $m_1 = m_2$

9.4 Distance Formulas

- ▶ Distance from point (x_1, y_1) to line $ax+by+c=0$: $d = \frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$
- ▶ Distance between parallel lines $ax+by+c_1=0$ and $ax+by+c_2=0$: $d = \frac{|c_1 - c_2|}{\sqrt{a^2+b^2}}$
- ▶ Foot of perpendicular from (x_1, y_1) to $ax+by+c=0$: use parametric method

9.5 General Equation of a Line

- ▶ $ax + by + c = 0$ is general form (a, b not both zero)
- ▶ Convert to slope-intercept: $y = (-a/b)x - c/b \Rightarrow \text{slope} = -a/b, \text{y-intercept} = -c/b$
- ▶ Family of lines through intersection of $l_1=0$ and $l_2=0$: $l_1 + \lambda l_2 = 0$ (λ any constant)

⚠ Common Mistakes

- Perpendicular slope: if $m=2$, perpendicular slope = $-1/2$ (negative reciprocal, NOT -2).
- Distance formula: use $|ax_1+by_1+c|/\sqrt{(a^2+b^2)}$, the absolute value is essential.
- Horizontal line $y=k$ has slope 0. Vertical line $x=k$ has undefined (not infinite) slope.
- Intercept form $x/a+y/b=1$: a and b are intercepts on x and y axes, not coordinates of a point.

➡ Worked Example

Q: Find the distance between parallel lines $3x-4y+7=0$ and $3x-4y-3=0$.

Step 1: Lines are parallel: same coefficients of x and y

Step 2: $a=3, b=-4, c_1=7, c_2=-3$

Step 3: $d = |c_1 - c_2|/\sqrt{(a^2+b^2)} = |7 - (-3)|/\sqrt{(9+16)} = 10/\sqrt{25} = 10/5 = 2$

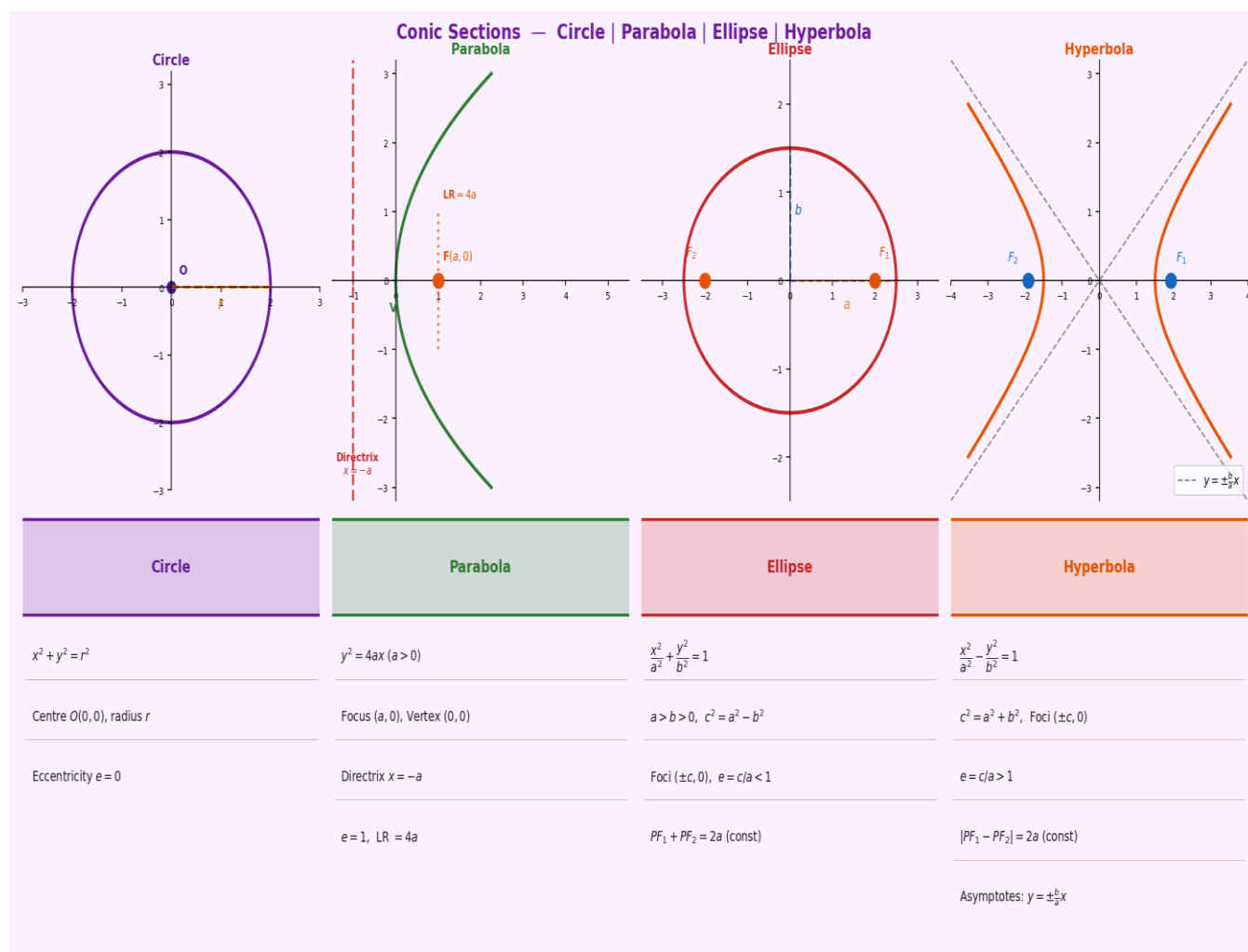
∴ Answer: Distance = 2 units

Type/Year	Question
SA	Find the equation of a line passing through $(2,3)$ with slope -4 .
SA	Find the angle between lines $3x-y+5=0$ and $x+3y-2=0$.
LA	Find the equation of the line through $(4,-3)$ perpendicular to $3x+5y-2=0$.
SA	Show that points $(1,1), (2,3), (3,5)$ are collinear.
SA	Find distance from $(3,4)$ to line $5x-2y-6=0$.

CHAPTER 10 · Unit III — Part of 12 Marks

Conic Sections

Circle | Parabola | Ellipse | Hyperbola



Graph 10.1: Four conic sections — Circle, Parabola, Ellipse, Hyperbola — with standard equations and key elements

10.1 Circle

- **Standard equation (centre origin):** $x^2 + y^2 = r^2$
- **General form (centre (h,k)):** $(x - h)^2 + (y - k)^2 = r^2$
- **General equation:** $x^2 + y^2 + 2gx + 2fy + c = 0$ | Centre $= (-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$

10.2 Parabola

Form	Equation	Focus	Directrix	Axis	Opens
Standard $y^2=4ax$	$y^2 = 4ax \ (a > 0)$	$(a, 0)$	$x = -a$	x-axis	Right
$y^2=-4ax$	$y^2 = -4ax \ (a > 0)$	$(-a, 0)$	$x = a$	x-axis	Left

$x^2=4ay$	$x^2 = 4ay$ ($a>0$)	(0, a)	$y = -a$	y-axis	Up
$x^2=-4ay$	$x^2 = -4ay$ ($a>0$)	(0, -a)	$y = a$	y-axis	Down

- Latus rectum of parabola = $4a$ (chord through focus perpendicular to axis)

10.3 Ellipse

- **Standard form ($a>b>0$):** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Property	Formula/Value
Foci	$\pm c$ where $c^2=a^2-b^2$ (on major axis)
Vertices	$(\pm a, 0)$ on major axis $(0, \pm b)$ on minor axis
Eccentricity e	$e = c/a$ where $0 < e < 1$
Latus rectum	Length = $2b^2/a$
Sum of focal distances	$PF_1 + PF_2 = 2a$ (constant for any point P on ellipse)

10.4 Hyperbola

- **Standard form:** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Property	Formula/Value
Foci	$\pm c$ where $c^2=a^2+b^2$
Vertices	$(\pm a, 0)$
Eccentricity e	$e = c/a$ where $e > 1$
Asymptotes	$y = \pm(b/a)x$
Difference of focal distances	$ PF_1 - PF_2 = 2a$



Conic Comparison at a Glance

- Circle: $e=0$ | Parabola: $e=1$ | Ellipse: $0<e<1$ | Hyperbola: $e>1$
- Parabola: one focus, one directrix | Ellipse & Hyperbola: two foci
- Ellipse: $PF_1 + PF_2 = 2a$ (sum constant) | Hyperbola: $|PF_1 - PF_2| = 2a$ (difference constant)
- The eccentricity e completely characterises the shape of a conic

⚠ Common Mistakes

- For ellipse: $c^2 = a^2 - b^2$ (subtract). For hyperbola: $c^2 = a^2 + b^2$ (add).

- Always check which axis is the major axis (larger denominator in ellipse).
- $y^2=4ax$ opens RIGHT, $y^2=-4ax$ opens LEFT. Sign of $4a$ determines direction.
- Asymptotes of hyperbola: $y=\pm(b/a)x$. They are NOT part of the hyperbola.

→ Worked Example

Q: Find the focus, vertex, directrix and latus rectum of $y^2=8x$.

Step 1: Compare with $y^2=4ax$: $4a=8 \Rightarrow a=2$

Step 2: Focus: $(a,0) = (2,0)$

Step 3: Vertex: $(0,0)$

Step 4: Directrix: $x = -a = -2$

Step 5: Latus rectum: $4a = 8$

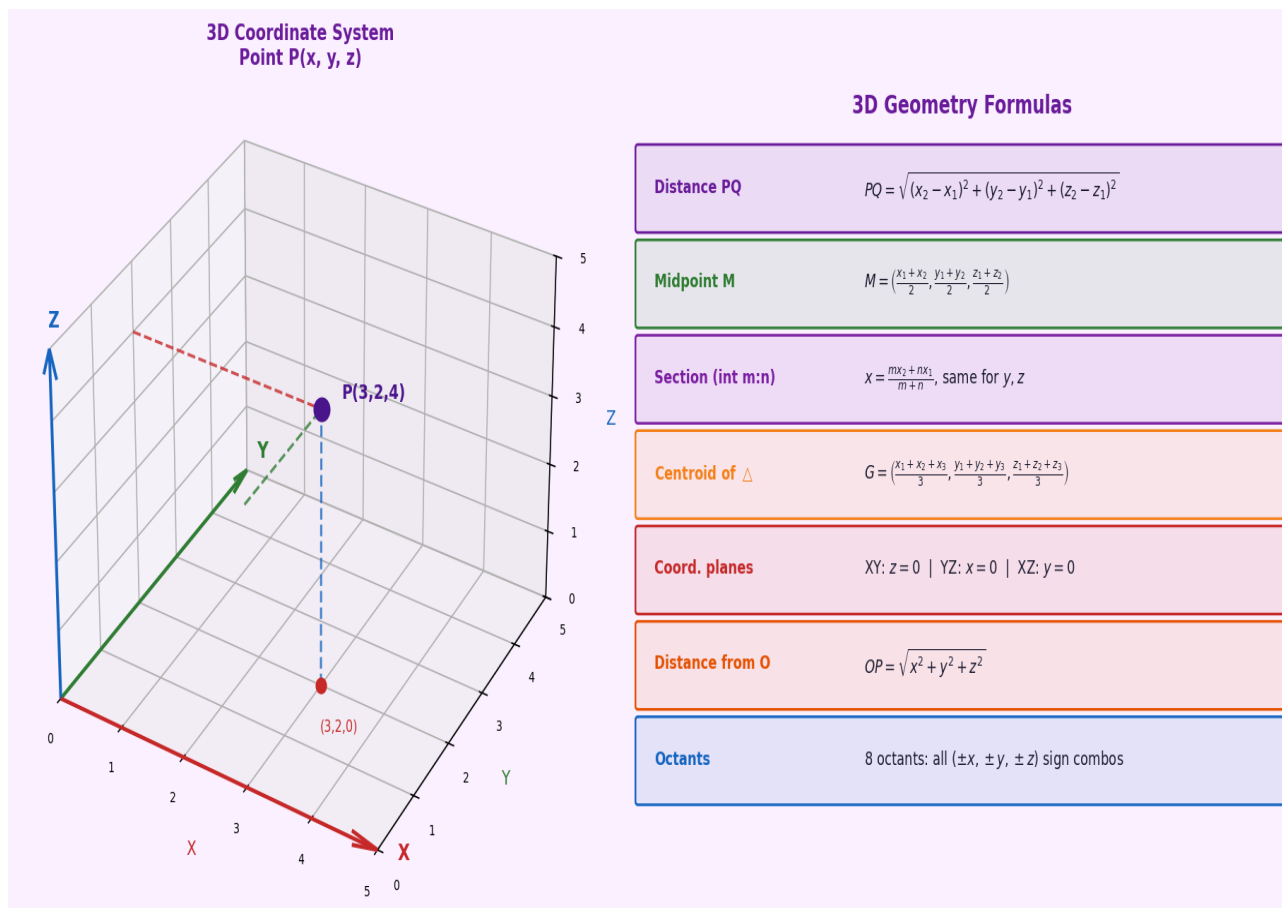
∴ Answer: Focus $(2,0)$, Vertex $(0,0)$, Directrix $x=-2$, Latus rectum = 8

Type/Year	Question
SA	Find the centre and radius of $x^2+y^2-6x+4y-3=0$.
SA	Find the equation of ellipse with foci at $(\pm 2,0)$ and $a=3$.
LA	Find the equation of parabola with vertex at origin, axis along y-axis, passing through $(5,2)$.
SA	For hyperbola $x^2/9-y^2/16=1$, find c , e and asymptotes.
SA	Find the equation of circle with centre $(3,-1)$ passing through $(5,2)$.

CHAPTER 11 · Unit III — Part of 12 Marks

Introduction to 3D Geometry

Coordinates | Distance | Section Formula | Octants



Graph 11.1: 3D coordinate system with point P(3,2,4) and projections (left) | 3D formula reference (right)

11.1 3D Coordinate System

- ▶ Three mutually perpendicular axes: X-axis, Y-axis, Z-axis meet at origin O(0,0,0)
- ▶ A point P(x,y,z): x=abscissa, y=ordinate, z=applycate
- ▶ XY-plane: $z=0$ | YZ-plane: $x=0$ | XZ-plane: $y=0$
- ▶ Eight octants defined by signs of (x,y,z): e.g. (+,+,+) is first octant

11.2 Distance Formula in 3D

- ▶ $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ distance between P(x₁,y₁,z₁) and Q(x₂,y₂,z₂)
- ▶ Distance from origin O(0,0,0) to P(x,y,z) = $\sqrt{x^2 + y^2 + z^2}$

11.3 Section Formula in 3D

- ▶ Point dividing PQ in ratio m:n (internal): $R = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$
- ▶ Midpoint of PQ: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$

- External division (ratio $m:n$): replace n with $-n$ in the formula

11.4 Centroid of Triangle in 3D

- Centroid $G = ((x_1+x_2+x_3)/3, (y_1+y_2+y_3)/3, (z_1+z_2+z_3)/3)$

⚠ Common Mistakes

- 3D distance: SQUARE all three differences, sum them, THEN take square root.
- Section formula: $m:n$ means m parts from first point, n from second. Don't reverse.
- Midpoint is a special case of section formula with $m=n=1$.
- Octant identification: all 8 combinations of $(+/-)$ signs for (x,y,z) .

➡ Worked Example

Q: Find the point that divides the line joining $A(2,-1,3)$ and $B(4,3,1)$ in ratio $2:1$ internally.

Step 1: $m=2, n=1; A=(2,-1,3), B=(4,3,1)$

Step 2: $x = (m \times x_2 + n \times x_1)/(m+n) = (2 \times 4 + 1 \times 2)/(2+1) = (8+2)/3 = 10/3$

Step 3: $y = (2 \times 3 + 1 \times (-1))/3 = (6-1)/3 = 5/3$

Step 4: $z = (2 \times 1 + 1 \times 3)/3 = (2+3)/3 = 5/3$

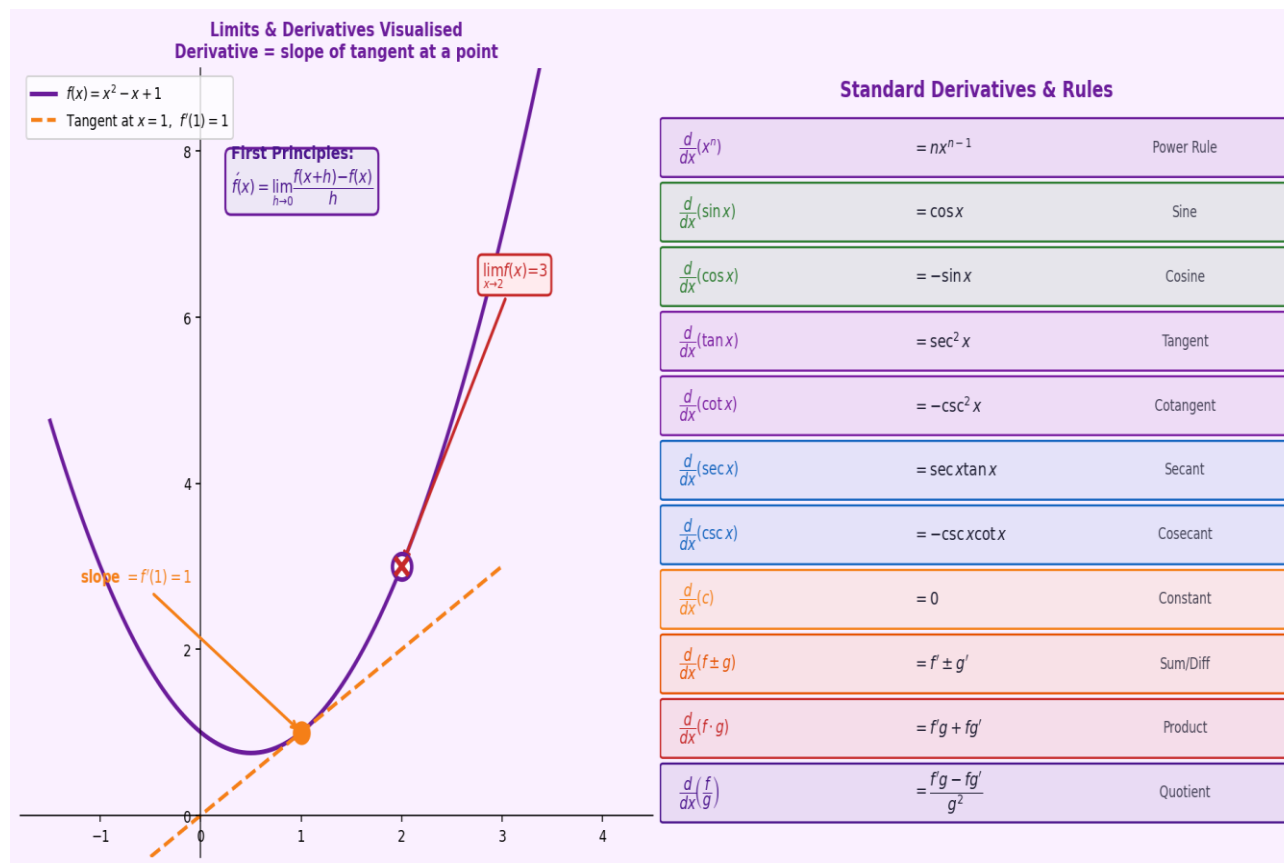
∴ Answer: Required point = $(10/3, 5/3, 5/3)$

Type/Year	Question
SA	Find the distance between points $A(3,4,-1)$ and $B(0,0,0)$.
SA	Find the midpoint of the segment joining $(2,3,1)$ and $(4,-1,3)$.
SA	Locate the point $(-2,3,-5)$ in the appropriate octant.
LA	Verify by distance formula that $A(1,2,3), B(4,0,4), C(2,-2,1)$ form a right triangle.
SA	Find the point on z -axis equidistant from $(1,2,3)$ and $(4,5,6)$.

CHAPTER 12 · Unit IV — 8 Marks

Limits and Derivatives

Limit Laws | Standard Limits | First Principles | Differentiation Rules

Graph 12.1: Limit of $f(x)$ as $x \rightarrow 2$ visualised (left) | Standard derivatives and differentiation rules (right)

12.1 Concept of Limit

- ▶ $\lim_{x \rightarrow a} f(x) = L$ means $f(x)$ approaches L as x approaches a (but $x \neq a$)
- ▶ Left-hand limit (LHL): $\lim_{x \rightarrow a^-} f(x)$ | Right-hand limit (RHL): $\lim_{x \rightarrow a^+} f(x)$
- ▶ Limit exists $\Leftrightarrow$ LHL = RHL. If they differ, limit doesn't exist at that point.

12.2 Standard Limits ★

- ▶ $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n \times a^{n-1}$ *polynomial limit identity*
- ▶ $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (in radians)
- ▶ $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$ (in radians)
- ▶ $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = n$

12.3 Limit Laws (Algebra of Limits)

- ▶ $\lim[f(x) \pm g(x)] = \lim f(x) \pm \lim g(x)$ (sum/difference)
- ▶ $\lim[f(x) \times g(x)] = \lim f(x) \times \lim g(x)$ (product)
- ▶ $\lim[f(x)/g(x)] = \lim f(x) / \lim g(x)$ provided $\lim g(x) \neq 0$
- ▶ $\lim[c \times f(x)] = c \times \lim f(x)$ (scalar multiple)

12.4 Derivatives — First Principles

- ▶ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ *definition of derivative (first principles)*
- ▶ Geometric meaning: derivative = slope of tangent to curve $y=f(x)$ at point $(x, f(x))$

12.5 Standard Derivatives & Rules ★

- ▶ $d/dx(x^n) = n \times x^{n-1}$ (**Power rule**) $d/dx(c) = 0$ (constant)
- ▶ $d/dx(\sin x) = \cos x$ | $d/dx(\cos x) = -\sin x$
- ▶ $d/dx(\tan x) = \sec^2 x$ | $d/dx(\cot x) = -\operatorname{cosec}^2 x$
- ▶ $d/dx(\sec x) = \sec x \tan x$ | $d/dx(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

Rules of Differentiation:

- ▶ **Product rule:** $d/dx(fg) = f'g + fg'$
- ▶ **Quotient rule:** $d/dx\left(\frac{f}{g}\right) = \frac{f'g - fg'}{g^2}$
- ▶ **Chain rule:** if $y=f(g(x))$, then $dy/dx = f'(g(x)) \times g'(x)$ [not in Class 11 syllabus][for reference]

⚠ Common Mistakes

- $\lim_{x \rightarrow 0} (\sin x)/x = 1$ ONLY when x is in radians. In degrees: answer is $\pi/180$.
- $d/dx(x^n) = nx^{n-1}$ NOT nx^n . The power reduces by 1.
- Product rule: $(fg)' = f'g + fg'$, NOT $f'g'$. Both terms are needed.
- In limit $h \rightarrow 0$: substitute $h=0$ directly if no $0/0$ form. Factor if $0/0$ indeterminate.

➡ Worked Example

Q: Find the derivative of $f(x) = 3x^2 + 5x - 7$ from first principles.

Step 1: $f(x+h) = 3(x+h)^2 + 5(x+h) - 7 = 3x^2 + 6xh + 3h^2 + 5x + 5h - 7$

Step 2: $f(x+h) - f(x) = 6xh + 3h^2 + 5h = h(6x + 3h + 5)$

Step 3: $[f(x+h) - f(x)]/h = 6x + 3h + 5$

Step 4: Limit as $h \rightarrow 0$: $f'(x) = 6x + 5$

∴ Answer: $f'(x) = 6x + 5$

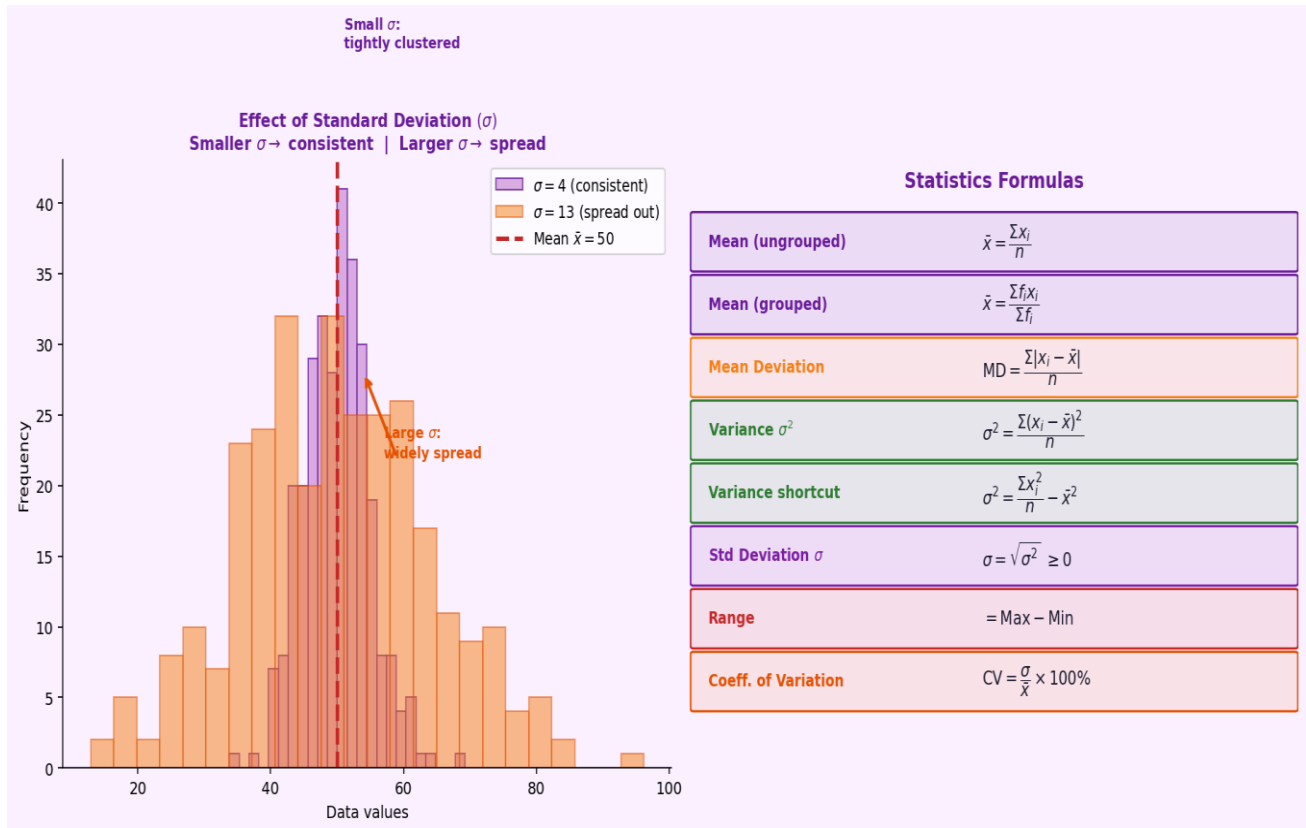
Type/Year	Question
SA	Find $\lim_{x \rightarrow 2} (x^2 - 4)/(x - 2)$.
SA	Differentiate $f(x) = x^3 + 7x^2 - 4x + 9$ with respect to x .

LA	Find derivative of $\sin x$ from first principles.
SA	If $y = (x^2+1)(x-3)$, find dy/dx using product rule.
SA	Find derivative of $(\sin x)/(1+\cos x)$.

CHAPTER 13 · Unit V — Part of 12 Marks

Statistics

Mean Deviation | Variance | Standard Deviation | Coefficient of Variation



Graph 13.1: Histograms showing small vs large standard deviation (left) | Statistics formula reference (right)

13.1 Measures of Central Tendency (Review)

- Mean (ungrouped): $\bar{x} = \frac{\sum x_i}{n}$
- Mean (grouped): $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ f_i = frequency, x_i = class mark

13.2 Mean Deviation

- MD (about mean): $\frac{\sum |x_i - \bar{x}|}{n}$ for ungrouped data
- MD (grouped): $\frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$ for frequency distribution

13.3 Variance and Standard Deviation ★

- Variance $\sigma^2 = \frac{\sum |x_i - \bar{x}|^2}{n}$ ungrouped
- Shortcut Variance $= \frac{\sum x_i^2}{n} - (\bar{x})^2$ faster computation

- ▶ **SD** $\sigma = \sqrt{\sigma^2}$ *standard deviation* = $\sqrt{(\text{variance})}$
- ▶ Coefficient of Variation (CV) = $(\sigma/\bar{x}) \times 100\%$ → smaller CV = more consistent/uniform

13.4 Comparing Distributions

🔗 Using CV to Compare Consistency

- $CV = (\text{SD}/\text{Mean}) \times 100\%$
- The distribution with **SMALLER CV** is more consistent/uniform
- CV is dimensionless so useful to compare data with different units or different means
- Example: Test A has mean 70, SD 10 $\Rightarrow CV=14.3\%$ | Test B has mean 40, SD 8 $\Rightarrow CV=20\%$. Test A is more consistent.

⚠ Common Mistakes

- Variance is ALWAYS ≥ 0 . Negative variance indicates a calculation error.
- $SD = \sqrt{(\text{Variance})}$. Don't forget the square root step.
- Mean deviation uses absolute values $|x - \bar{x}|$. Never ignore the absolute value.
- For comparing consistency, use CV not raw SD (especially when means differ).

➡ Worked Example

Q: Find variance and SD of: 4, 7, 13, 2, 1, 3

Step 1: $n=6$. Sum = $4+7+13+2+1+3 = 30$. Mean $\bar{x} = 30/6 = 5$

Step 2: Deviations from mean: $-1, 2, 8, -3, -4, -2$

Step 3: $\sum(x_i - \bar{x})^2 = 1+4+64+9+16+4 = 98$

Step 4: $\sigma^2 = 98/6 = 49/3 \approx 16.33$

Step 5: $\sigma = \sqrt{49/3} = 7/\sqrt{3} \approx 4.04$

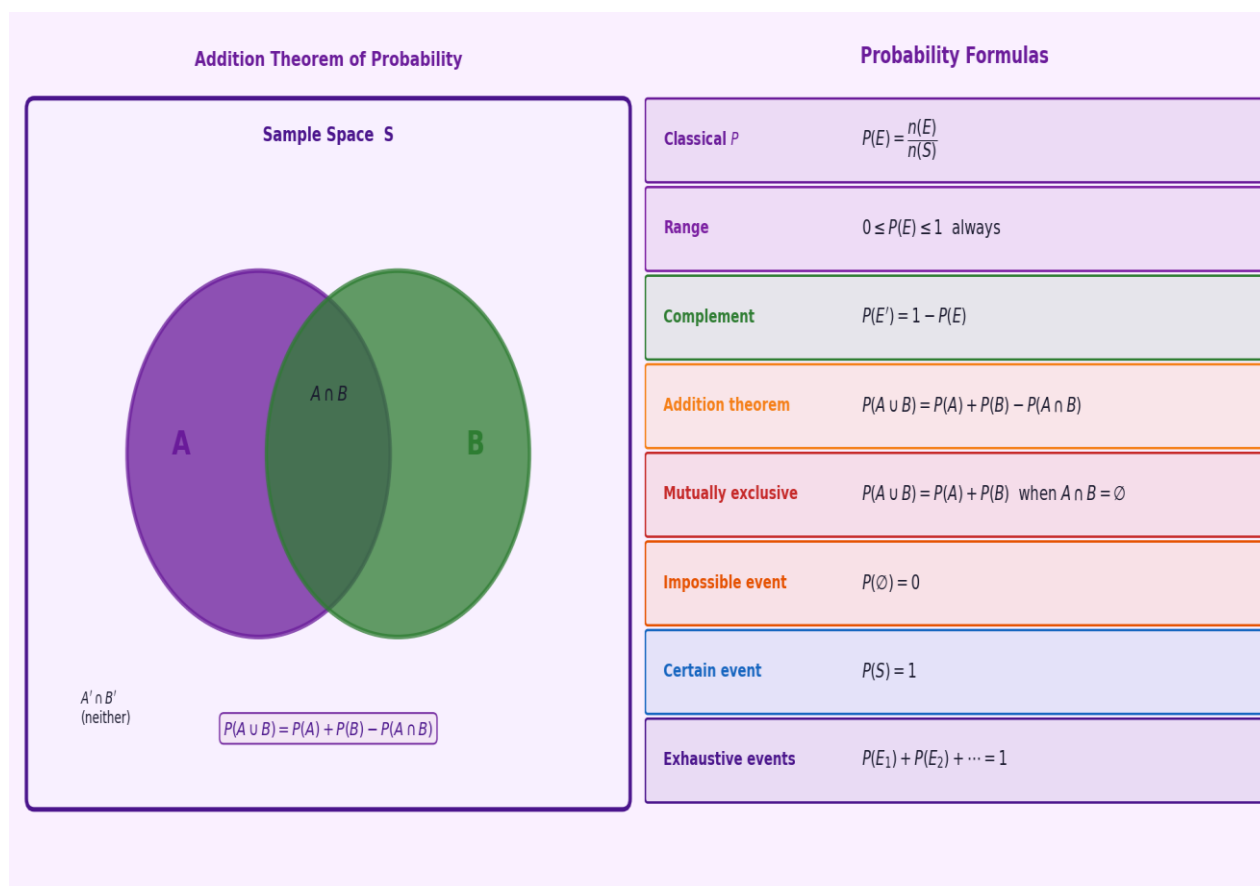
∴ Answer: $\sigma^2 = 49/3$, $\sigma = 7/\sqrt{3} \approx 4.04$

Type/Year	Question
SA	Find mean deviation about mean for: 6, 7, 10, 12, 13, 4, 8, 12.
SA	Which is more consistent? A: mean=60, SD=5 or B: mean=40, SD=6?
LA	Find variance and SD of first 10 natural numbers.
SA	If variance of x_i is 4, find variance of $(2x_i+3)$.
SA	Frequency table: $x: 2,4,6,8,10; f: 1,2,3,2,1$. Find mean and variance.

CHAPTER 14 · Unit V — Part of 12 Marks

Probability

Sample Space | Events | Classical | Addition Theorem

Graph 14.1: Venn diagram for Addition Theorem $P(A \cup B)$ (left) | Key probability formulas reference (right)

14.1 Basic Terminology

Term	Definition	Example
Experiment	A process that produces a well-defined outcome	Tossing a coin
Sample Space (S)	Set of all possible outcomes	{H, T} for coin toss
Event	A subset of the sample space	{H} = getting heads
Sure/Certain event	S itself (always occurs)	Getting a number on a die
Impossible event	$\emptyset$ (never occurs)	Getting 7 on a standard die
Complementary A'	All outcomes NOT in A	$A = \{H\} \Rightarrow A' = \{T\}$

Mutually exclusive	$A \cap B = \emptyset$ (cannot both occur simultaneously)	Even and odd on one die roll
Exhaustive	$A \cup B \dots = S$ (together cover everything)	Even and odd cover all die outcomes

14.2 Classical (Theoretical) Probability

- ▶ $P(E) = \frac{n(E)}{n(S)}$ *number of favourable outcomes ÷ total equally likely outcomes*
- ▶ $0 \leq P(E) \leq 1$ for any event E
- ▶ $P(S) = 1 \mid P(\emptyset) = 0$
- ▶ $P(A') = 1 - P(A)$ (complement rule)

14.3 Addition Theorem of Probability ★

- ▶ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ [General Addition Rule]
- ▶ $P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive ($A \cap B = \emptyset$)
- ▶ $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

14.4 Common Sample Spaces

Experiment	Sample Space n(S)	Key Facts
Tossing 1 coin	$S = \{H, T\}$, $n=2$	$P(H) = P(T) = 1/2$
Tossing 2 coins	$S = \{HH, HT, TH, TT\}$, $n=4$	$P(2 \text{ heads}) = 1/4$, $P(\text{at least 1 head}) = 3/4$
Tossing n coins	$n(S) = 2^n$	$P(\text{all heads}) = 1/2^n$
Rolling 1 die	$S = \{1, 2, 3, 4, 5, 6\}$, $n=6$	$P(\text{even}) = P(\text{odd}) = 1/2$
Rolling 2 dice	$n(S) = 36$	$P(\text{sum}=7) = 6/36 = 1/6$ (most likely sum)
Deck of cards	$n(S) = 52$	13 each: spades, hearts, diamonds, clubs 4 suits

🔗 Card Facts (52-card deck)

- 4 suits: Spades♠(black), Hearts♥(red), Diamonds♦(red), Clubs♣(black)
- 13 cards per suit: A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K
- Face cards (picture cards): J, Q, K $\Rightarrow 3 \times 4 = 12$ face cards
- Honours/Court cards: A, J, Q, K $\Rightarrow 4 \times 4 = 16$
- $P(\text{king}) = 4/52 = 1/13 \mid P(\text{red card}) = 26/52 = 1/2$

⚠ Common Mistakes

- $P(A \cup B) \neq P(A) + P(B)$ unless A and B are mutually exclusive.

- $P(A') = 1 - P(A)$, not $-P(A)$. Probability is always between 0 and 1.
- Events can only be mutually exclusive OR exhaustive independently — check both conditions.
- When rolling two dice: $n(S) = 36$, not 12. Each die has 6 faces, and $6 \times 6 = 36$.

➡ Worked Example

Q: A card is drawn from 52 cards. Find $P(\text{king or red card})$.

Step 1: $P(\text{King}) = 4/52 = 1/13$

Step 2: $P(\text{Red}) = 26/52 = 1/2$

Step 3: $P(\text{King AND Red}) = 2/52 = 1/26$ (red kings: King of Hearts and King of Diamonds)

Step 4: $P(\text{King or Red}) = P(\text{King}) + P(\text{Red}) - P(\text{King and Red}) = 4/52 + 26/52 - 2/52 = 28/52 = 7/13$

∴ Answer: $P(\text{King or Red card}) = 7/13$

Type/Year	Question
SA	Two dice are thrown. Find $P(\text{sum}=8)$ and $P(\text{at least one } 4)$.
SA	From a deck, a card is drawn. Find $P(\text{queen or black card})$.
LA	In a class of 60: 25 play football, 20 play cricket, 10 play both. Find $P(\text{plays at least one})$ if a student is chosen randomly.
SA	Three coins tossed. Find $P(\text{at least 2 heads})$.
SA	Events A and B: $P(A)=1/2$, $P(B)=1/3$, $P(A \cap B)=1/6$. Are A and B mutually exclusive?

Master Quick Revision Sheet — All 14 Chapters (2025-26)

Chapter	★ Must-Know Formulas & Key Points
Ch 1 Sets	De Morgan: $(A \cup B)' = A' \cap B'$ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ Power set: 2^n subsets
Ch 2 Relations	Domain/range $f(x)/g(x)$: exclude zeros $f \circ g$ means $f(g(x))$ One-one: $f(a)=f(b) \Rightarrow a=b$
Ch 3 Trig	$\sin^2 + \cos^2 = 1$, $\sec^2 - \tan^2 = 1$, $\operatorname{cosec}^2 - \cot^2 = 1$ $\sin 2A = 2 \sin A \cos A$ $\tan \pm B$ formula ASTC
Ch 4 Complex	$i^2 = -1$ $ z = \sqrt{a^2 + b^2}$ Division: multiply by conjugate Polar: $r(\cos \theta + i \sin \theta)$
Ch 5 Inequalities	Flip direction when multiplying/dividing by negative Open circle for strict Shade feasible region
Ch 6 P&C	$nPr = n!/(n-r)!$ $nCr = n!/[r!(n-r)!]$ Circular $= (n-1)!$ $0! = 1$ Pascal: $nCr + nC(r-1) = (n+1)Cr$
Ch 7 Binomial	$T(r+1) = nCr \times a^{n-r} \times b^r$ Middle term: n even $\rightarrow 1$ term; n odd $\rightarrow 2$ terms $\sum nCr = 2^n$
Ch 8 Sequences	AP: $a_n = a + (n-1)d$, $S_n = n/2[2a + (n-1)d]$ GP: $a_n = ar^{n-1}$, $S_\infty = a/(1-r)$ $AM \geq GM \geq HM$
Ch 9 Lines	Slope $m = \tan \theta$ Distance $= (ax_1 + by_1 + c)/\sqrt{a^2 + b^2}$ Perp: $m_1 m_2 = -1$ Parallel: $m_1 = m_2$
Ch 10 Conics	Circle: $(x-h)^2 + (y-k)^2 = r^2$ Parabola: $y^2 = 4ax$ Ellipse: $c^2 = a^2 - b^2$ Hyperbola: $c^2 = a^2 + b^2$
Ch 11 3D	Distance: $\sqrt{[\sum (x_i - x_j)^2]}$ Midpoint = avg of coordinates 8 octants $c^2 = a^2 + b^2$ (3D Pythagoras)
Ch 12 Calculus	$\lim (\sin x)/x = 1$ $d/dx(x^n) = nx^{n-1}$ Product: $(fg)' = f'g + fg'$ Quotient: $(f/g)' = (f'g - fg')/g^2$
Ch 13 Statistics	$MD = \sum x_i - \bar{x} /n$ $\sigma^2 = \sum (x_i - \bar{x})^2/n$ $\sigma = \sqrt{\sigma^2}$ $CV = (\sigma/\bar{x}) \times 100\%$ Smaller CV = more consistent
Ch 14 Probability	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $P(A') = 1 - P(A)$ Card deck: 52, 4 suits, 13 each $n(S)$ for 2 dice = 36

Board Exam Strategy 2025-26 (80 Marks Theory + 20 Internal)

Topic	Strategy
High Yield Units	Algebra (25M) $\Rightarrow$ Sets & Trig (23M) $\Rightarrow$ Stats & Prob (12M) $\Rightarrow$ Coord Geo (12M) $\Rightarrow$ Calculus (8M)
Trigonometry	Memorise ALL identities: Pythagorean, compound angle, double angle, product-to-sum
Conics	Know all 4 conics: focus, directrix, eccentricity, latus rectum, equation for

	each
P & C	Decision: Does order matter? YES $\Rightarrow nPr$, NO $\Rightarrow nCr$. Circular $= (n-1)!$
Limits	For 0/0 form: factorise then cancel Use standard limits as direct formulas
Derivatives	Memorise power/sin/cos/tan. Apply product rule and quotient rule step-by-step
Statistics	Write all formulas first, then compute step-by-step. Show all working
Probability	List sample space, count favourable outcomes. Use Addition Theorem correctly
Proof Questions	Write: Given, To Prove, Proof. Every step must be justified
Internal 20M	5M pen-paper test + 5M portfolio + 5M multiple assessment + 5M lab/activities

Final Words

- This book covers 100% of the official CBSE 2025-26 Class 11 Maths syllabus.
- Algebra (Unit II, 25 marks) is the highest-weightage unit — master all 5 chapters!
- Trig identities and conic sections appear almost every year in LA questions.
- For full video solutions and past papers, visit [sjmaths.com](https://www.sjmaths.com)
- Practice — not just reading — is what turns concepts into marks. All the best! 